

BANAK LAKSELV, Norway

WMO: 010590

Lat: 70.069N

Lon: 24.973E

Elev: 25

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.4	-6.1	-16.8	2.2	-11.1	-11.6	2.9	-5.8	40.6	25.9	36.6	24.6	5.9	180	0.826

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	73.2	59.4	69.0	57.7	65.0	55.9	61.2	70.3	58.9	66.9	56.8	63.7	12.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	69.5	65.2	55.0	64.5	62.0	53.2	60.4	60.2	27.2	70.8	25.7	67.0	24.4	63.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.6	28.7	25.1	-16.1	80.8	5.9	5.3	-20.3	84.6	-23.8	87.7	-27.1	90.7	-31.3	94.6		
(5)			-16.6	64.9	5.7	3.1	-20.7	67.1	-24.1	68.9	-27.3	70.7	-31.4	72.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	16.9	16.6	23.2	32.1	41.3	49.1	55.3	52.6	46.2	35.9	25.5	21.9
	DBStd	15.85	11.85	12.10	9.64	7.27	6.51	5.88	6.22	5.60	6.09	7.48	8.95	10.43
	HDD50	5936	1025	935	831	537	282	84	15	35	145	440	735	872
	HDD65	11025	1490	1355	1296	987	736	477	308	387	564	903	1185	1337
	CDD50	397	0	0	0	0	11	58	181	114	31	2	0	0
	CDD65	10	0	0	0	0	0	0	8	2	0	0	0	0
	CDH74	106	0	0	0	0	0	4	80	22	0	0	0	0
	CDH80	22	0	0	0	0	0	0	19	3	0	0	0	0
Wind	WSAvg	11.2	13.7	13.5	12.9	10.4	10.5	10.0	8.9	8.4	9.5	11.0	12.3	13.2
Precipitation	PrecAvg	18.4	1.3	1.1	1.1	1.0	1.2	1.8	2.2	2.3	1.9	1.9	1.3	1.4
	PrecMax	22.6	2.4	2.0	2.0	1.8	3.3	3.1	5.2	6.1	3.7	3.5	2.7	2.8
	PrecMin	12.6	0.3	0.2	0.2	0.3	0.3	0.3	0.4	1.1	0.5	0.8	0.2	0.5
	PrecStd	2.6	0.6	0.5	0.5	0.4	0.7	0.8	1.2	1.1	0.7	0.7	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	43.2	45.0	52.9	68.4	74.2	82.6	77.5	66.2	55.1	46.1	45.8
		MCWB	37.8	38.2	39.1	43.7	52.5	57.2	64.3	61.9	55.8	48.6	41.6	40.5
	2%	DB	38.9	40.3	41.2	48.4	61.0	68.3	75.2	71.0	62.3	51.6	41.6	41.1
		MCWB	35.0	35.7	36.2	41.4	49.5	54.9	60.9	59.0	54.2	46.2	38.0	36.6
	5%	DB	35.5	36.1	37.9	44.7	55.6	63.4	70.4	66.4	58.9	48.5	38.9	37.8
		MCWB	32.2	32.6	33.6	39.1	46.3	53.2	59.2	57.1	52.4	43.8	35.7	34.2
	10%	DB	32.3	32.4	35.5	42.0	51.6	59.0	66.4	62.7	55.7	45.7	36.6	35.4
		MCWB	29.8	29.8	31.7	36.9	44.2	50.5	57.6	55.4	50.3	42.0	33.9	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.1	39.1	39.4	44.6	53.8	60.2	66.2	63.8	57.7	49.2	41.9	40.5
		MCDB	42.2	43.2	45.0	51.1	65.9	68.7	78.4	73.3	64.0	54.0	45.3	45.2
	2%	WB	34.9	35.8	36.5	41.7	49.6	56.7	62.6	60.2	55.0	46.7	38.1	36.9
		MCDB	38.5	40.1	40.9	47.6	59.9	65.8	73.0	68.5	60.6	51.2	41.1	40.6
	5%	WB	32.3	32.9	34.0	39.5	47.0	53.8	60.0	58.1	53.0	44.2	35.9	34.6
		MCDB	35.1	36.0	37.6	44.4	54.6	61.9	68.7	65.2	58.2	47.4	38.4	37.8
	10%	WB	29.7	30.0	31.8	37.2	44.7	51.3	57.8	56.0	50.9	42.1	34.0	32.4
		MCDB	32.2	32.4	35.2	41.2	50.7	58.0	65.3	61.9	55.2	45.2	36.2	35.1
Mean Daily Temperature Range	5% DB	MDBR	12.1	11.9	12.7	12.3	12.1	11.8	13.3	13.8	11.7	8.5	10.0	11.4
		MCDBR	13.3	13.2	10.6	13.8	19.6	20.2	22.0	21.5	16.4	10.5	10.5	11.9
		MCWBR	11.2	11.0	8.4	8.8	10.7	10.2	10.5	11.1	9.9	7.4	8.7	9.6
	5% WB	MCDBR	13.1	13.0	10.6	12.8	18.5	18.2	20.2	19.8	14.9	9.7	9.9	11.9
		MCWBR	11.2	11.0	8.7	8.4	10.6	10.2	10.6	11.1	9.5	7.1	8.5	9.9
Clear-Sky Solar Irradiance	taub	N/A	0.250	0.302	0.327	0.345	0.345	0.355	0.338	0.308	0.260	N/A	N/A	
	taud	N/A	2.031	2.102	2.159	2.296	2.449	2.456	2.483	2.520	2.183	N/A	N/A	
	Ebn at Noon	0	142	216	252	264	269	261	252	228	140	0	0	
	Edh at Noon	0	13	25	33	33	29	28	24	17	11	0	0	
All-Sky Solar Radiation	RadAvg	0	97	498	1113	1505	1533	1388	949	485	141	0	0	
	RadStd	0	10	33	62	134	124	147	89	46	15	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)			+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page